

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourg Public Administration — Civil Servant NLU Deployment

Definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Justification

Benchmark inputs are drawn predominantly from RTL news articles, the Luxembourgish Online Dictionary, Wikipedia/Wikidata, and a translated English voice-assistant corpus [Q12, Q16, Q24, Q30, Q34, Q46, Q51]. The authors explicitly warn that 'most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts' [Q112], and that LTZ data 'often originate from a limited set of public domains' [Q119]. The deployment, however, expects orthographically variable, code-switched (Luxembourgish/French/German) citizen correspondence [elicitation Q2; regional context languages.code_switching_patterns]. Only the human-annotated NER sub-corpus from RTL user comments provides partial coverage of informal and code-mixed writing [Q41]. The intent detection corpus uses a voice-assistant register that has no LTZ reference, so terminology could not be systematically verified [Q55, Q56]. Given that ~47% of Luxembourg's workforce are frontaliers (mostly French-speaking) [WEB-5, WEB-2] and the language law requires the administration to respond in the citizen's chosen language [WEB-14], the benchmark's content distribution is materially narrower than the deployment input distribution. IC is HIGH priority and the documented gaps are substantial.

ID	Evidence
WEB-5	~231,290 frontaliers (~47% of employment) — citizen-input distribution will be heavily multilingual
WEB-2	OECD 2025: 50% of frontaliers from France, remainder split between Belgium and Germany — supports differentiated administrative-query profile
WEB-14	Language law requires response in citizen's chosen language — implies language-of-input output dimension absent from benchmark

Strengths

- Lothritz et al. NER sub-corpus from RTL online news comments provides partial informal and code-mixed writing coverage [Q40, Q41]
- Authors transparently document content limitations, register narrowness, and demographic-coverage caveats [Q110, Q112, Q119, Q121]
- Pre-training corpus for LTZ-E1 includes broader registers (user comments, webchat, transcribed podcasts and political speeches) [Q74], which may benefit downstream model representations even if task evaluation is narrower

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — citizen correspondence requires handling of orthographically variable Luxembourgish, French/German code-switching, frontier-specific terminology, and Luxembourg-specific institutional vocabulary [elicitation Q2; regional context languages.code_switching_patterns, cultural_norms_notes]. The benchmark inputs are dominated by RTL news and

dictionary content [Q12, Q16, Q46], which do not encode this distribution.

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IC-2: Partially. Sources are Luxembourgish in origin so cultural framing is locally grounded, but the institutional/journalistic register diverges from citizen-government correspondence [Q112]. The voice-assistant intent corpus is translated from English without LTZ register reference [Q55, Q56], introducing alien register material.

IC-3: Yes — the xSID-derived intent corpus (alarm/reminder/weather) [Q51, Q55] is Western voice-assistant content with no administrative analogue in Luxembourg; its inputs encode construct-irrelevant register. Authors note pre-training data overlap with LLMs may also inflate performance [Q111].

IC-4: INSUFFICIENT DOCUMENTATION — no Luxembourg civil-administration domain experts appear to have reviewed task inputs for representativeness; the elicitation flags this as needing real-time civil-servant feedback for progressive alignment [elicitation Q3].

IC-5: Documented content-validity issues: (a) formal/institutional register dominance [Q112], (b) absence of systematic code-switching coverage outside NER [Q41], (c) voice-assistant register import without LTZ reference [Q55, Q56], (d) no representation of frontier-specific administrative content [WEB-5].

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Information Gaps

- No quantification of code-switching frequency in any task split
- No measure of orthographic-variation coverage in task inputs
- Whether RTL user-comment NER sub-corpus contains enough French/German switching to represent deployment distribution

Requires Expert Verification

- Sampling and qualitative review of LTZGLUE task inputs by civil-administration linguists to confirm representativeness for citizen-correspondence content

Remediation

Gap 1: Benchmark inputs are dominated by formal/institutional registers and filter out non-Luxembourgish text; code-switching and orthographic variation are under-represented

Recommendation: Construct an internal evaluation corpus from anonymized real citizen correspondence (with CNPD-compliant DPIA) that explicitly preserves French/German code-switching, frontier terminology, and orthographic variation. Use it as an out-of-distribution stress test alongside LTZGLUE.